ParentSync

Your family's WhatsApp and email messages, automatically organized into calendar events

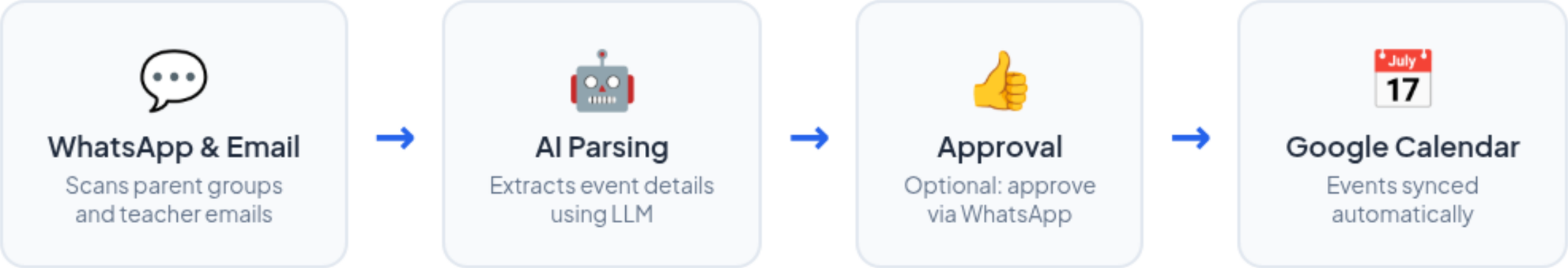
v1.0.1 | Desktop App for Linux, Windows & macOS

What is ParentSync?

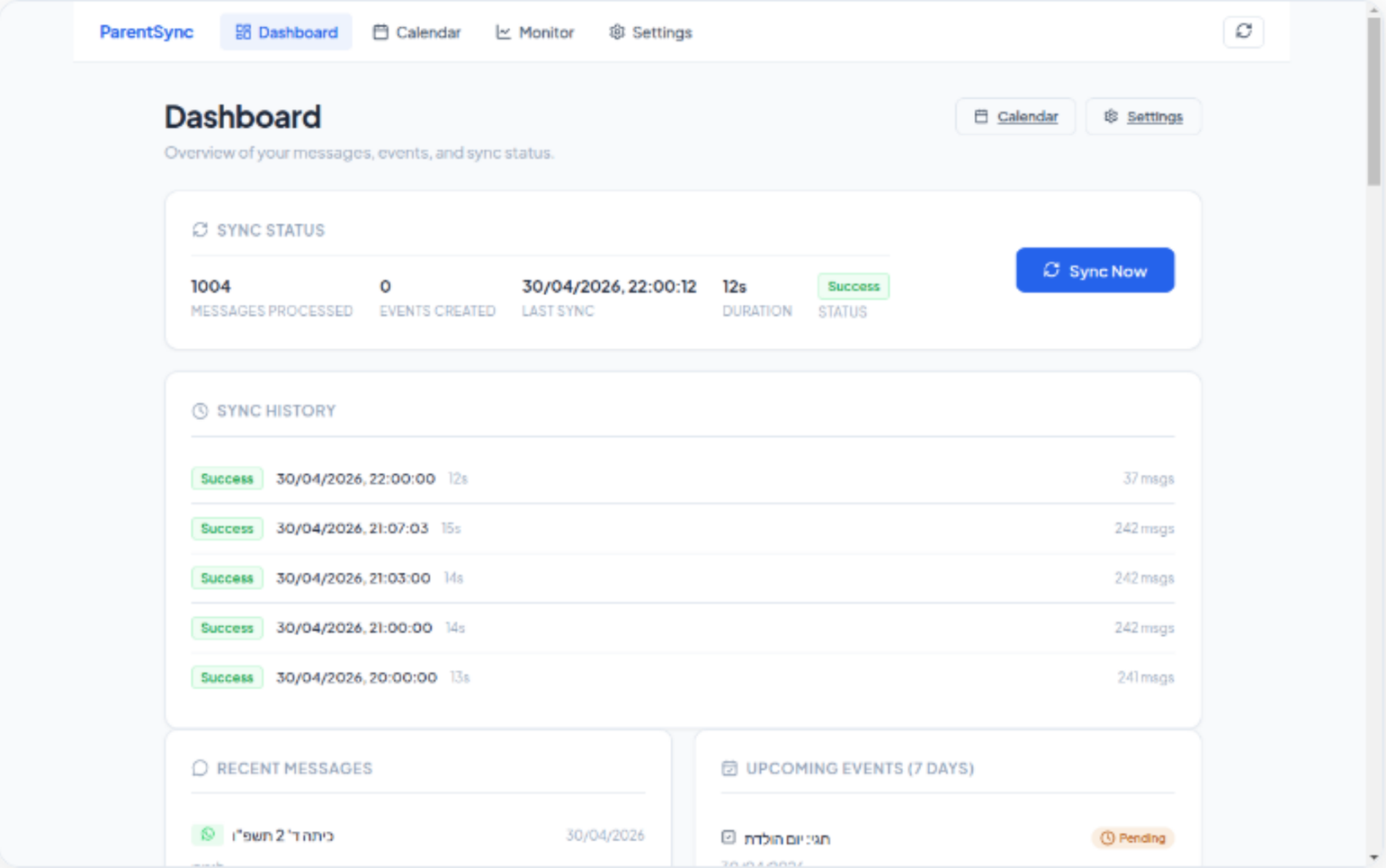
A desktop app that reads your children's school WhatsApp groups and teacher emails, uses AI to find events, and puts them on your Google Calendar.

- Monitors WhatsApp parent groups and Gmail automatically
- AI-powered parsing extracts dates, times, and event details
- Events sync directly to your family Google Calendar
- Optional approval workflow via WhatsApp reactions (👍 approve / ☹️ reject)
- All data stored locally on your machine — no cloud, no sign-up

How It Works

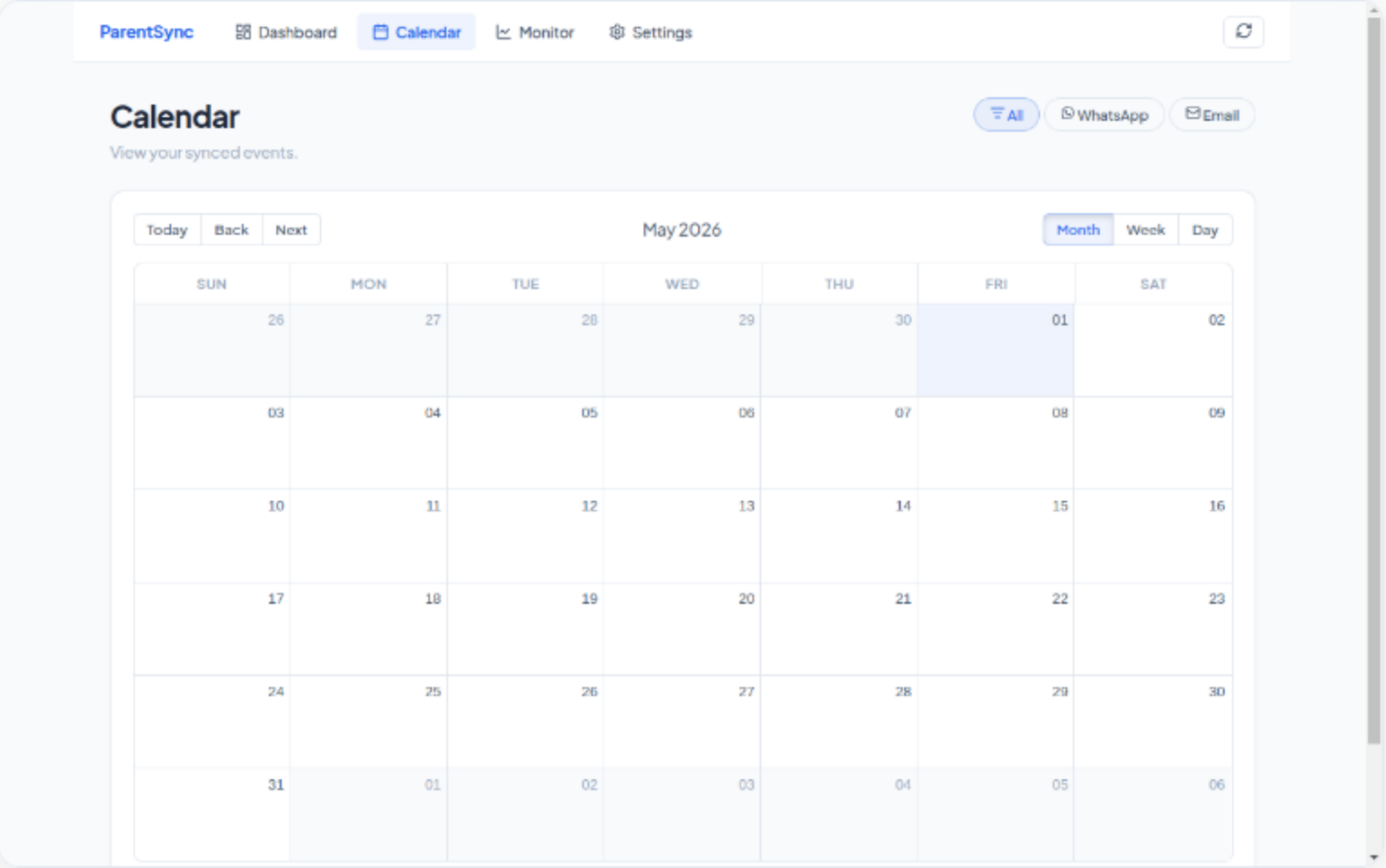

WhatsApp & Email
Scans parent groups and teacher emails
AI Parsing
Extracts event details using LLM
Approval
Optional: approve via WhatsApp
Google Calendar
Events synced automatically

Dashboard



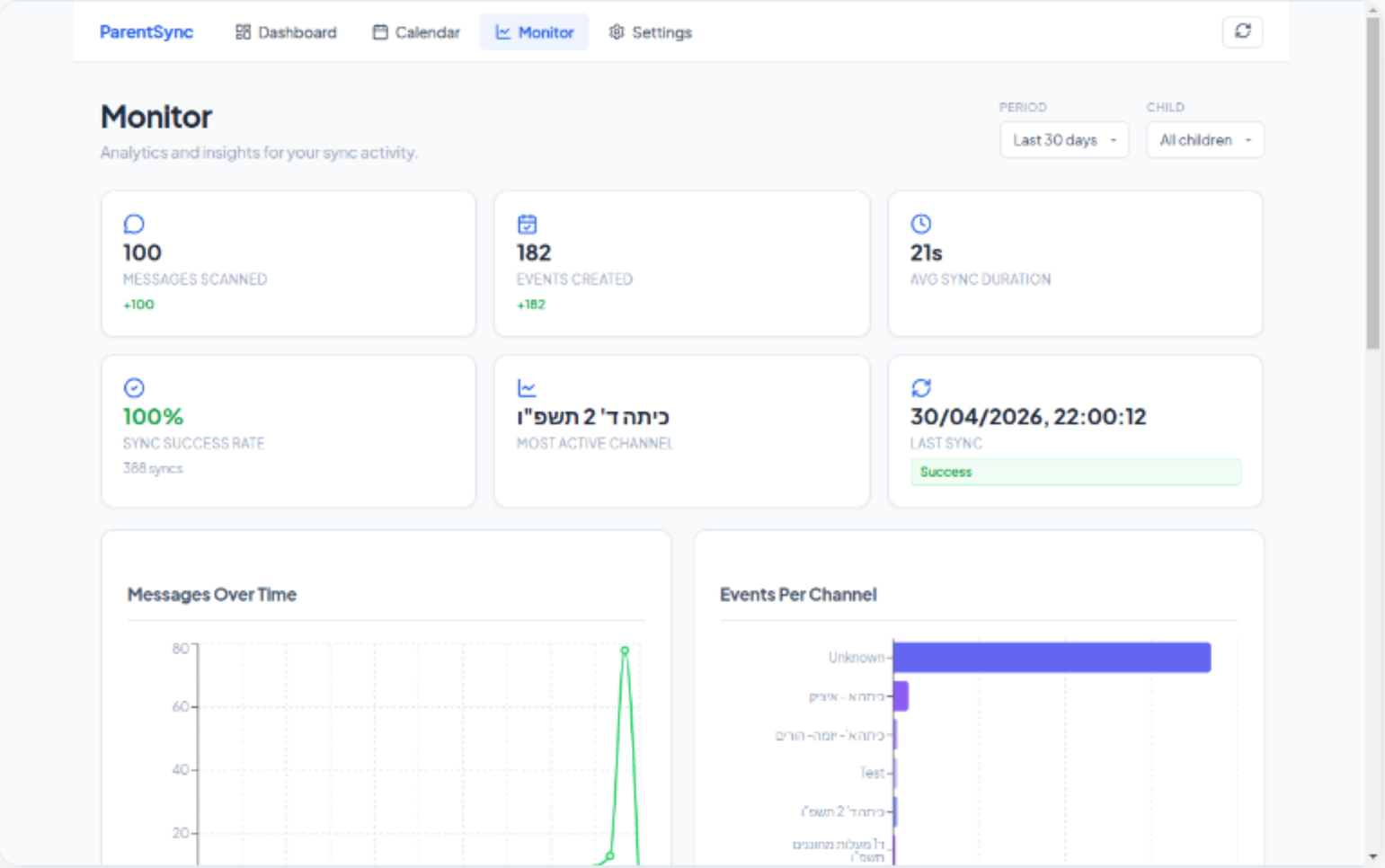
Sync status, recent messages, upcoming events, and sync history at a glance

Calendar View



Full month/week/day calendar with events color-coded by source (WhatsApp or Email)

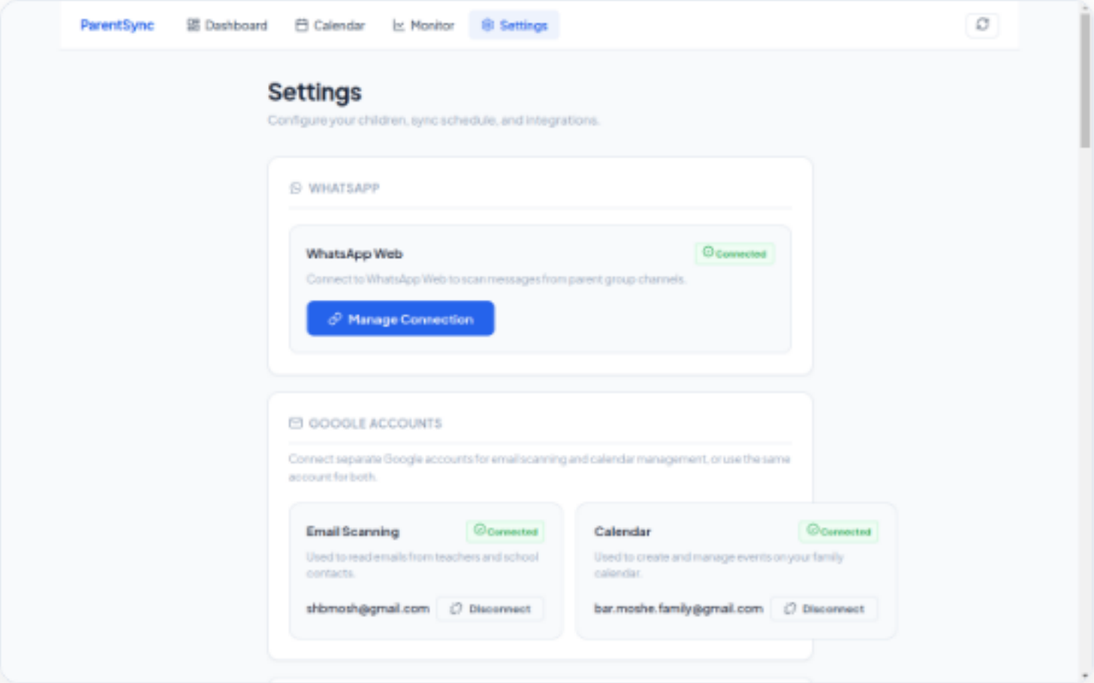
Monitor & Analytics



Messages scanned, events created, sync success rate, and channel activity charts

Settings & Setup

Connections



Children & Schedule

Dashboard — In Detail

Sync Status (top card)

- **Messages Processed** — total messages collected across all recent syncs
- **Events Created** — how many calendar events the AI extracted
- **Last Sync** — when the last sync happened
- **Duration** — how long it took (e.g., 18s)
- **Status badge** — green "Success", orange "Partial", or red "Failed"

💡 Click **"Sync Now"** to trigger an immediate scan of all your configured WhatsApp groups and email addresses.

Sync History (expandable list)

- Shows the **last 5 syncs** with status, time, duration, message count
- **Click any entry** to expand — see per-channel breakdown (which child, which group, how many messages)
- Skipped channels show the **reason** (e.g., "no new messages")
- Click **"View synced messages"** to see exact messages that were collected

Dashboard — Messages & Events

Recent Messages (bottom left)

- All messages collected from WhatsApp and Gmail
- Source icon: **WhatsApp icon** = WhatsApp group, **Mail icon** = email
- **Click a message** to expand and read full content
- Click **"Show more"** to load additional messages (20 at a time)

Upcoming Events (bottom right)

- Events in the **next 7 days** parsed from messages
- Coloured pill: 🟠 **Pending**, 🟢 **Approved**, 🔴 **Rejected**
- Pending events have inline **Approve / Reject** buttons — same effect as 👍 / ☹ in WhatsApp
- **Checkmark icon** – synced to Google Calendar

How to get data flowing

1. Make sure **WhatsApp is connected** (Settings → WhatsApp → "Connected")
2. Add at least one **child with WhatsApp channels** (Settings → Children)
3. Set your **Gemini API key** (Settings → Gemini AI), or use OpenRouter as alternative
4. Click **"Sync Now"** on the Dashboard
5. Wait 15–30 seconds — messages and events appear

💡 **Refresh button** (top-right ↻ icon) reloads all data on the current page without triggering a new sync.

Monitor — Summary Cards

Six cards at the top give you a quick health check. Use the **Period** dropdown (7/30/90 days) and **Child** filter to narrow the view.

💬 Messages Scanned

Total WhatsApp + email messages collected. Green trend arrow shows change vs. previous period.

⏱️ Avg Sync Duration

Average time per sync cycle (scanning + parsing). Under 30s is normal.

📊 Most Active Channel

The WhatsApp group or email source that generated the most messages in this period.

📅 Events Created

How many calendar events the AI extracted from messages. Trend arrow compares to previous period.

✅ Sync Success Rate

Percentage of syncs that completed without errors. Green ≥80%, orange ≥50%, red below 50%.

🔄 Last Sync

Timestamp and status badge of the most recent sync. Tells you when data was last refreshed.

Monitor — Charts & Graphs

📈 Messages Over Time

Line chart with two lines:

🟢 **Green** = WhatsApp messages

🔴 **Red** = Email messages

X-axis: dates, Y-axis: message count. Shows daily volume trends — helps spot when groups are most active.

📊 Events Per Channel

Horizontal bar chart showing event count by WhatsApp group name. Each channel gets a different purple shade. Hover to see percentage of total. Identifies which groups generate the most calendar events.

📉 Sync History

Combo chart — bars + line overlay.

Bars: message count per sync (green = success, orange = partial, red = failed)

Line: sync duration in seconds (right axis)

Spot slow syncs or recurring failures at a glance.

🗓️ Channel Activity Heatmap

Grid with channels as rows, dates as columns. Cell color intensity = message count. Darker purple = more messages. Hover any cell for exact count. Reveals which days and channels are busiest.

Common Errors & Solutions

⚠️ "Sync failed"

WhatsApp may be disconnected, or OpenRouter key is missing/expired. Check Settings → WhatsApp status and OpenRouter API key.

⚠️ "0 Events Created" after sync

The AI didn't find any events in the messages. This is normal if messages are just chat. Only messages with dates/times get parsed into events.

⚠️ "Google OAuth not configured"

You haven't entered the Google Client ID and Secret yet. Go to Settings → Google OAuth and follow the setup steps (slide 16).

⚠️ "WhatsApp is not connected"

Session expired. Go to Settings → click "Connect WhatsApp" → scan QR code again with your phone.

⚠️ "Failed to load dashboard data"

Backend may still be starting. Wait a few seconds and click the **refresh button** (↻ top-right). If it persists, restart the app.

⚠️ Events not appearing on Google Calendar

Check: 1) Google Calendar account is connected (Settings), 2) If using approval channel, react 👍 to approve the event in WhatsApp.

Getting & Refreshing Data

Automatic Sync

ParentSync automatically scans at the hours you configured in Settings → Sync Schedule. For example, if you selected 8am–10pm, it scans once per hour during those times.

Manual Sync

Click **"Sync Now"** on the Dashboard to trigger an immediate scan. Useful after adding a new child or channel.

Refresh Display

The **↻ refresh button** (top-right corner of the nav bar) reloads all data on the current page from the database. It does NOT trigger a new sync — it just refreshes the view.

Data Freshness Tips

1. After connecting WhatsApp for the first time, click **"Sync Now"** to populate initial data
2. Add all your children's group names **before** the first sync so all channels are scanned
3. The Monitor page filters (7d/30d/90d) affect all charts — use **30 days** for a good overview
4. If Dashboard shows old data, click **↻ refresh** — if it still doesn't update, click **"Sync Now"**

💡 The app keeps running and syncing even when minimized to the system tray.

How-To Guide

Step-by-step setup for each integration

Setup: Gemini (default AI)

The AI engine that reads messages and extracts calendar events

What is Gemini?

Google's Gemini family of LLMs — fast, cheap, and very good at multilingual extraction. ParentSync uses it to read WhatsApp and email messages and identify dates, events, and details.

💡 The default model `gemini-2.5-flash-lite` is fast and cheap — typical extraction costs are negligible. A free tier is available too.

🔁 Don't want Gemini? **OpenRouter** is supported as an alternative provider — see next slide.

How to get your API key

1. Go to **aistudio.google.com/apikey**
2. Sign in with your Google account
3. Click **"Create API key"**
4. Copy the key
5. Paste in ParentSync → Settings → Gemini AI → API Key
6. Optionally pick a different model in the same section

Alternative: OpenRouter

One-stop shop for many LLM providers behind a single API key

What is OpenRouter?

A gateway to many AI models (GPT, Claude, Llama, Qwen, etc.) through a single API key. Useful if you already have credits there, or if you want to A/B test models without changing keys.

💡 Free models are available — typically labelled `:free`. They have rate limits but cost nothing.

How to get your API key

1. Go to **openrouter.ai/keys**
2. Sign up or log in (Google/GitHub)
3. Click **"Create Key"**
4. Copy the key (starts with `sk-or-`)
5. Paste in ParentSync → Settings → OpenRouter → API Key

You can fill both keys; the active provider is whichever the build is wired to (Gemini by default).

Setup: Google Account

Connect Gmail and Google Calendar to read emails and sync events

What is Google OAuth?

OAuth lets ParentSync access your Gmail and Calendar **without storing your Google password**. You create a "Client ID" in Google Cloud Console — think of it as a permission slip that tells Google "this app is allowed to access my data."

💡 You can use **separate Google accounts** for email scanning and calendar — or the same account for both.

How to set it up

1. Go to **console.cloud.google.com**
   Create a project if you don't have one
2. Enable **Gmail API** and **Calendar API**
   APIs & Services → Enable APIs
3. Create **OAuth consent screen**
   Add your email as a Test user
4. Create **OAuth Client ID** (Web app)
   Redirect URI:
   `http://localhost:41932/api/auth/google/callback`
5. Copy **Client ID** and **Client Secret** into ParentSync Settings

Connecting Google Accounts

After entering Client ID & Secret

1. Click **"Save Settings"**
2. Click **"Sign in with Google"** under Email Scanning
3. Google's consent screen opens in your browser
4. Select your account and click **"Allow"**
5. You're redirected back — status shows **Connected**
6. Repeat for **Calendar** (same or different account)

Troubleshooting

⚠️ **"Error 403: access_denied"**

Your Google app is in "Testing" mode. Fix: Go to OAuth consent screen → either **add your email as a Test user** or click **"Publish App"**.

⚠️ **"Failed to exchange authorization code"**

The Client Secret may be wrong, or the redirect URI doesn't match. Double-check both in Google Cloud Console and in ParentSync Settings.

Setup: WhatsApp

Connect your WhatsApp to scan parent group messages

How it works

ParentSync connects to WhatsApp Web — the same way you'd use WhatsApp on your computer. It runs its own browser in the background to read messages from the groups you specify.

💡 Your WhatsApp session **persists across app restarts**. You only need to scan the QR code once. If the session expires, you'll be prompted to re-scan.

📱 Your phone must stay connected to the internet for WhatsApp Web to work (WhatsApp requirement).

How to connect

1. Open ParentSync → **Settings**
2. Click **"Connect WhatsApp"**
3. A **QR code** appears in the app
4. On your phone:
   WhatsApp → Settings → Linked Devices → Link a Device
5. Scan the QR code with your phone camera
6. Status changes to **"Connected"**

Then go to **Children** in Settings, add a child, and type the exact WhatsApp group name.

Under the Hood

DESKTOP SHELL
Electron (single executable)

BACKEND
NestJS (REST API, embedded)

AI
Gemini (default) · OpenRouter swap-in

FRONTEND
React + TypeScript

DATABASE
SQLite (local, no server)

WHATSAPP
whatsapp-web.js (Chromium)

All data stored locally on your machine. No cloud servers, no accounts to create. Just download, configure, and run.

Platform Support

Platform	Format	How to Run
Linux	.AppImage (portable)	chmod +x, double-click or run from terminal
Linux	.deb (Debian/Ubuntu)	dpkg -i, run from app menu
Windows	.exe (installer)	Double-click installer
macOS	.dmg	Drag to Applications

```
# Linux quick start
chmod +x ParentSync-1.0.1.AppImage
./ParentSync-1.0.1.AppImage
```

Built with Electron | All data stored locally | Private use

Event Reminders

ParentSync sends a WhatsApp reminder ~24 hours before an event begins, but only for events that have settled in the system long enough to be trusted.

- Hourly cron scans the calendar for upcoming events
- Only events added more than **24 hours ago** are eligible — protects against last-minute parsing mistakes
- Only events already **synced to Google Calendar** qualify
- Event existence is **verified against Google Calendar** before sending — deleted events are silently skipped
- Reminder is delivered to the configured approval/notification WhatsApp chat with full details: title, date, time, location, description, and source channel
- Each event is reminded at most once (`reminderSent` flag)

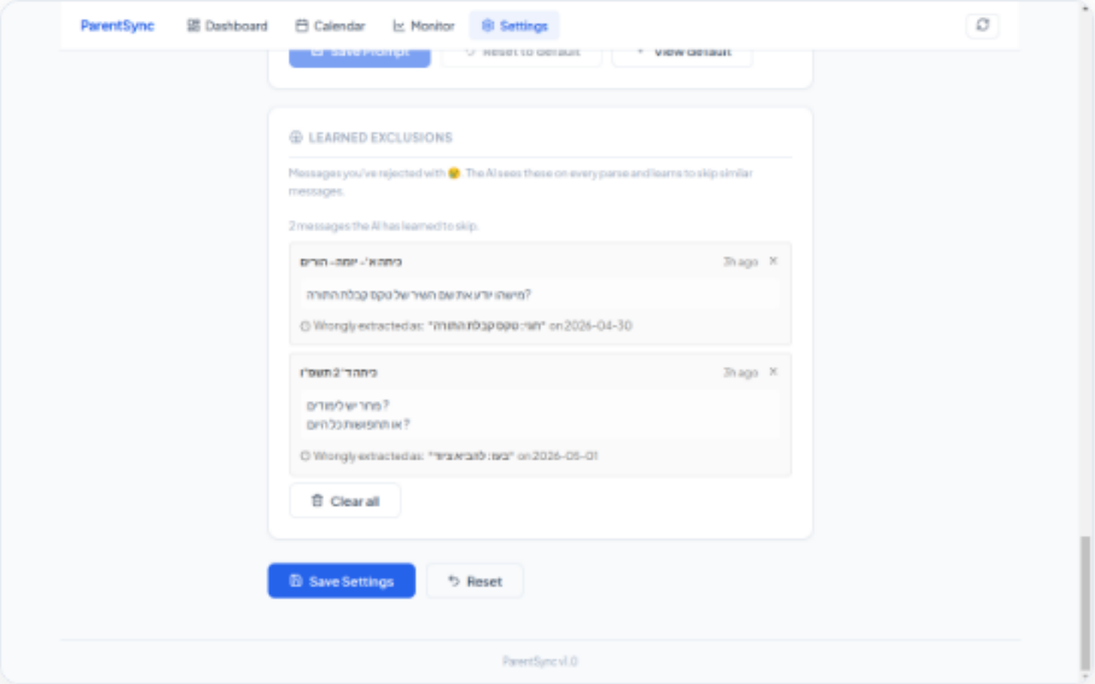
See `docs/EVENT-REMINDERS.md` for the full specification.

AI Feedback Loop

The system prompt that tells the LLM how to extract events is fully editable, and the app learns from every ☹ reaction.

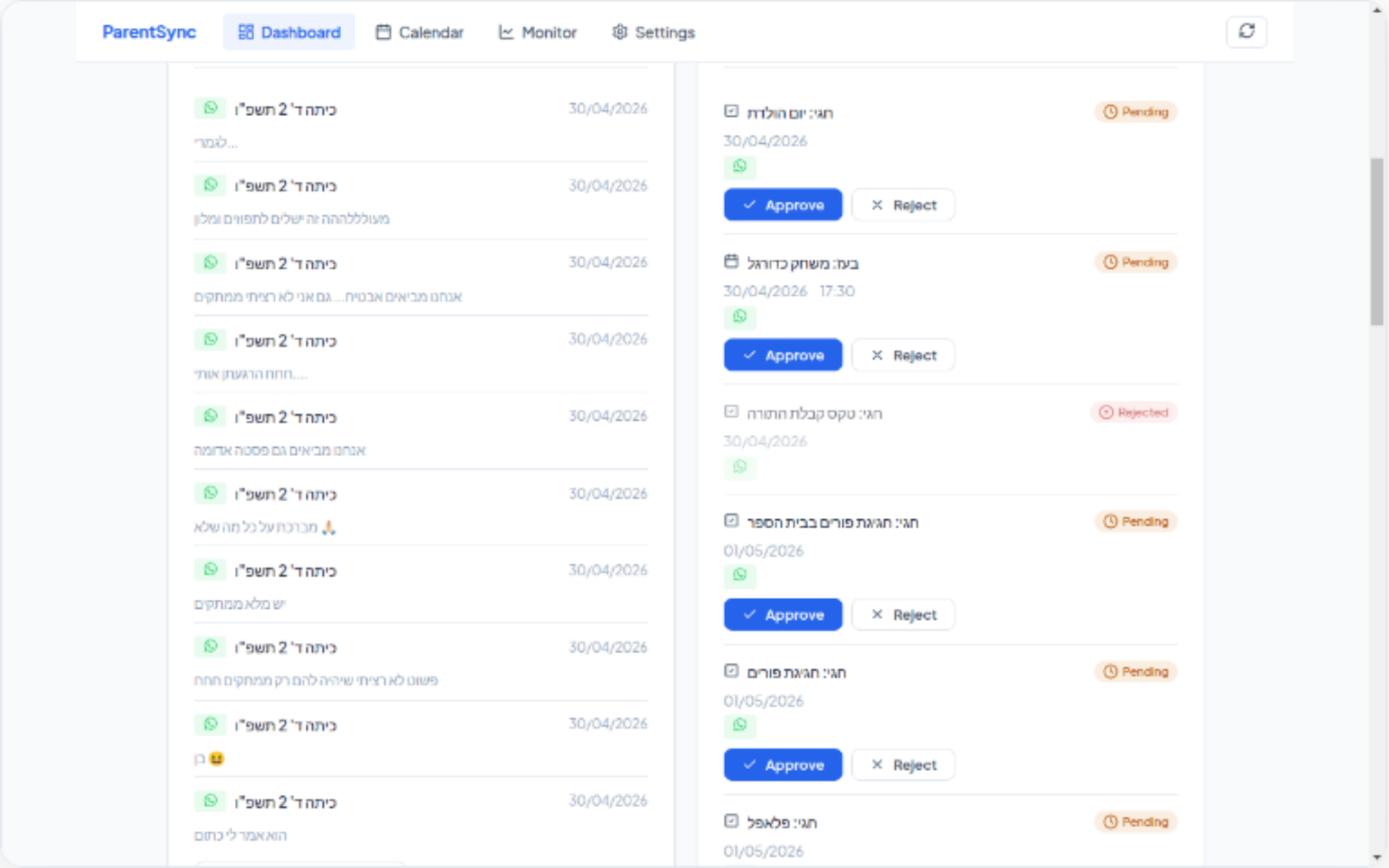
AI Extraction Prompt — editable from Settings

Learned Exclusions — captured from ☹ reactions



Every ☹ saves the source message + the wrong title. The most recent 50 are appended to the prompt as "do NOT extract events for messages similar to these," so the LLM stops repeating the mistake. The parse cache keys include a hash of the prompt + exclusions — so feedback closes on the next sync, not 24h later.

In-App Approval



Each upcoming event shows its status as a pill — 🟠 Pending, 🟢 Approved, 🔴 Rejected. Pending events have inline Approve / Reject buttons. Same effect as 👍 / ☹️ in WhatsApp; reactions are reversible too — take back a 👍 to unsync from Google, take back a ☹️ to clear the learned exclusion.

Duplicate Suppression

One real-world WhatsApp message can describe the same gathering from multiple angles — a birthday is also a "meetup," a class trip is also a "ceremony." The LLM used to extract them as *separate* events at the same date+time. Now:

- Before sending an event for approval, the backend looks up other non-rejected events for the same child at the same date+time slot.
- For each match, a focused LLM call asks **"are these two events the same gathering?"** The prompt is tiny and cheap.
- If the answer is yes, the new event is silently rejected and never reaches your approval channel — the existing event remains the canonical record.
- Date-only tasks (no time) skip the check entirely — the false-positive risk is too high without a time slot.
- LLM hiccups default to "different" — a transient error never accidentally swallows a real event.

💡 The check pairs with the negative-feedback loop: even if a duplicate slips through, one ☹ reaction puts it in the learned-exclusions pool so it's gone for good.